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DIRECT ADAPTIVE IMPEDANCE CONTROL OF MANIPULATORS

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Abstract
This paper presents an adaptive scheme for controlling the end-effector impedance of robot manipulators. The proposed control system consists of three subsystems: a simple "filter" which characterizes the desired dynamic relationship between the end-effector position error and the end-effector/environment contact force, an adaptive controller which produces the Cartesian-space control input required to provide this desired dynamic relationship, and an algorithm for mapping the Cartesian-space control input to a physically realizable joint-space control torque. The controller does not require knowledge of either the structure or the parameter values of the robot dynamics, and is implemented without calculation of the robot inverse kinematic transformation. As a result, the scheme represents a very general and computationally efficient approach to controlling the impedance of both nonredundant and redundant manipulators. Furthermore, the method can be applied directly to trajectory tracking in free-space motion by removing the impedance filter.

1. Introduction

Increasing the applicability and utility of robot manipulators requires the development of strategies for controlling the robot when it is in contact with the environment. Robots are subject to interaction forces whenever they perform tasks involving motion which is constrained by the environment, such as assembling parts or deburring edges. In such motions, the interaction forces must be accommodated rather than resisted, so as to comply with the environmental constraints.

Recognizing the importance of compliant motion for robots, many researchers have investigated this problem in recent years. As a result of these studies, two broad approaches to achieving compliant motion have emerged. The first approach, usually called ***hybrid position/force control***, is based on the observation that when the robot end-effector is in contact with the environment, the Cartesian-space of the end-effector coordinates may be naturally decomposed into a "position subspace" and a "force subspace" [1]. The position subspace and force subspace correspond to the Cartesian directions in which the end-effector is, respectively, free to move and constrained by the environment. The hybrid position/force control approach to compliant motion is to track a position (and orientation) trajectory in the position subspace and a force (and moment) trajectory in the force subspace.

The second approach to compliant motion for robots is called ***impedance control***, and proposes that the control objective should not be the tracking of position/force trajectories, but rather should involve the regulation of the mechanical impedance of the robot end-effector which relates position and force [2]. Thus the objective of the impedance controller is to maintain a desired dynamic relationship between the end-effector position and the end-effector/environ-ment contact force. This gives a unified framework for both unconstrained and constrained motion control problems and does not require the control switching required for hybrid control. In this paper, the impedance control approach to compliant motion for robots is considered.

Following the pioneering work of Hogan [2,3], several researchers have studied impedance control of robots and have demonstrated the potential of this approach to compliant motion [e.g., 4-10]. Much of this research is either conceptual in nature, or proposes control schemes designed using "classical" robot control techniques, such as the Computed Torque method [2-6]. The resulting controllers often require precise knowledge of the robot dynamic model and parameter values, as well as extensive computations related to the inverse kinematic transformation. Additionally, it is often difficult to ensure close tracking of fast trajectories with these controllers, or to achieve good performance in the presence of uncertainties, sensor noise, and external disturbances. Recent investigations have focused on enhancing the capabilities of robot impedance controllers, and have included efforts to develop adaptive impedance controllers [7-10].

This paper considers the problem of developing an adaptive impedance control scheme for manipulators. It is interesting to examine this problem from the perspective provided by the research performed on the adaptive unconstrained motion control problem. This topic has attracted considerable attention over the past decade, and no attempt will be made here to deal comprehensively with the vast literature in this area. Instead, the adaptive unconstrained motion control will be studied in terms of the representation of the robot dynamics utilized in the development of the control strategy. Within this framework, two broad classes of control algorithms can be identified. The first utilizes the following representation of the robot dynamic model:

$$\mathbf{T} = W(\theta,\dot{\theta},\ddot{\theta})\mathbf{p} \tag{1}$$

where $\mathbf{T} \in \Re^n$ is the joint torque vector, $\theta \in \Re^n$ is the joint coordinate vector, $\mathbf{p} \in \Re^p$ is the vector of unknown constant dynamic model parameters, and $W \in \Re^{n\times p}$ is the "regressor" matrix that embodies the structure of the robot dynamics and is assumed to be known. This "linear in the unknown parameters" model was first used in robot parameter estimation algorithms, and subsequently formed the basis for developing a class of globally stable algorithms for adaptive unconstrained motion control [e.g., 11,12]. Following [13], we shall call these schemes ***indirect adaptive controllers*** because the adaptive laws provide an explicit estimate of $\mathbf{p}$, which is then used in the control law. Although these controllers have performed well in both computer simulations and experiments, there are some potential difficulties associated with this approach. For example, implementation of these schemes with robots possessing more than two or three joints can be computationally intensive, particularly if it is desired to implement them directly in Cartesian-space. More seriously, their design requires precise knowledge of the structure of the entire robot dynamic model, including friction and transmission effects, which is often an unrealistic requirement in practice. Additionally, recent studies have indicated that these indirect adaptive controllers may lack robustness to unmodelled dynamics, sensor noise, and other disturbances [14,15].

The second approach to adaptive unconstrained motion control to be considered here will be termed ***direct adaptive control***

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[13] because the adaptive laws adjust the controller gains directly. These schemes typically use the following dynamic model representation in their development:

$$\mathbf{T} = H(\theta)\ddot{\theta} + \mathbf{V}(\theta,\dot{\theta}) + \mathbf{G}(\theta) \quad (2)$$

where $H \in \Re^{n\times n}$ is the symmetric positive-definite inertia matrix, $\mathbf{V} \in \Re^{n}$ represents the torques due to Coriolis and centripedal acceleration as well as all friction effects, and $\mathbf{G} \in \Re^{n}$ is the gravity torque vector. Typically the development of the direct adaptive controllers has proceeded by assuming that very little information is available concerning either the structure or the parameter values of the terms in (2); indeed, it is ordinarily only required to know that H in (2) is symmetric and positive-definite to develop the controller design [e.g, 16,17]. However, the control algorithm derivations rely on the assumption that the adaptive elements in the controller can vary significantly more rapidly that the elements in the dynamics (2), which has caused some concern regarding the stability of the resulting control systems in fast motions. Nevertheless, it must be noted that these direct adaptive controllers are extremely simple, computationally efficient, require very little model information, and have demonstrated good tracking performance and robustness characteristics in simulations and experiments with industrial six degree-of-freedom robots.

The research reported to date on the problem of adaptive compliant motion control fits naturally into the framework described above. The impedance controllers [7,8] are direct adaptive schemes developed in a manner similar to the one used for unconstrained motion control in [17], while the strategies [9,10] are indirect adaptive algorithms that represent extensions of the unconstrained motion controllers [11,12] to the constrained motion control problem. Thus it may be expected that the schemes [7-10] will inherit the virtues and the problems of the unconstrained motion controllers to which they are related, and that the problems may be compounded because of the interaction with the environment. The present paper proposes an adaptive impedance controller designed to overcome these difficulties. This controller is developed using a direct adaptive control methodology, and consists of three subsystems: a simple "filter", a Cartesian-space impedance controller, and a mapping algorithm. The filter provides a simple means for characterizing the desired dynamic relationship between the end-effector position error and the end-effector/environment contact force. The Cartesian-space impedance controller generates the control input required to provide this desired dynamic relationship. The controller is a direct adaptive strategy that does not require knowledge of either the structure or the parameter values of the robot dynamics. It is shown that the scheme is globally stable in the presence of bounded disturbances, and that the size of the tracking errors can be made arbitrarily small. The mapping algorithm transforms the Cartesian-space control input into a physically realizable joint control torque, and provides effective utilization of any available kinematic redundancy. The complete algorithm represents a very general and computationally efficient system for controlling the impedance of both nonredundant and redundant manipulators.

This paper is organized as follows. In Section 2 the impedance control problem is precisely stated, some preliminary facts are established, and the overall structure of the proposed controller is given. The Cartesian-space impedance control scheme is developed in Section 3, and the algorithm for mapping the control input generated by this scheme to a joint torque vector is summarized in Section 4. Finally, Section 5, summarizes the paper and draws some conclusions.

2. Problem Formulation

In this section, the impedance control problem is precisely stated in a form suitable for our subsequent analysis. Additionally, some relevant facts are established and the overall structure of the proposed controller is given. Attention is focused on n degree-of-freedom rigid link manipulators performing tasks in m-dimensional Cartesian-space, with $m \leq n$. This allows nonredundant and redundant robots to be studied together, and also permits convenient consideration of nonredundant manipulators involved in applications with a variable dimensional task-space.

Let $\mathbf{x} \in \Re^{m}$ define the position and orientation of the end-effector in Cartesian-space, and denote the relationship between $\mathbf{x}$ and θ as

$$\mathbf{x} = \mathbf{f}(\theta) \quad (3)$$

or

$$\dot{\mathbf{x}} = J(\theta)\dot{\theta} \quad (4)$$

where $\mathbf{f} : \Re^{n} \rightarrow \Re^{m}$ represents the forward kinematics of the robot and $J = \partial\mathbf{f}/\partial\theta \in \Re^{m\times n}$ is the end-effector Jacobian matrix. The impedance control objective may be concisely stated by specifying the desired dynamic relationship between the end-effector position $\mathbf{x}$, the end-effector reference trajectory $\mathbf{x}_r \in \Re^{m}$, and the vector of forces and torques exerted by the end-effector on the environment $\mathbf{P} \in \Re^{m}$ as follows:

$$M(\ddot{\mathbf{x}}_r - \ddot{\mathbf{x}}) + B(\dot{\mathbf{x}}_r - \dot{\mathbf{x}}) + N(\mathbf{x}_r - \mathbf{x}) = \mathbf{P} \quad (5)$$

where $\mathbf{x}_r$ is bounded and twice-differentiable. The matrices $M, B, N \in \Re^{m\times m}$ are specified by the user so that the dynamics (5) possesses the desired characteristics. For example, these matrices are often interpreted as the mass, damping, and stiffness of a mass-spring-damper system which quantify the mechanical impedance relationship between end-effector contact force and end-effector position error. It can be seen from (5) that, in the absence of environmental contact, we have $\mathbf{P} = \mathbf{0}$ and $\mathbf{x}(t) = \mathbf{x}_r(t) \ \ \forall t$ (assuming $\mathbf{x} = \mathbf{x}_r$ initially), which corresponds to free-space trajectory tracking. In the presence of contact the desired system response is for the end-effector to comply with the interaction forces in a manner similar to a spring-mass-damper system. Ordinarily the matrices M, B, N are diagonal with positive constants on the diagonals, in which case the dynamical system (5) is very simple. However, there is considerable freedom in choosing these matrices, and this flexibility is one of the attractive features of the impedance control approach to compliant motion.

As the impedance control objective is stated in terms of Cartesian-space variables, it is natural to study the impedance control problem directly in Cartesian-space. Consider the robot dynamic model in task-space

$$\mathbf{F} = H_{\mathbf{x}}(\mathbf{x})\ddot{\mathbf{x}} + D_{\mathbf{x}}(\mathbf{x})[\dot{\mathbf{x}}\dot{\mathbf{x}}] + \mathbf{G}_{\mathbf{x}}(\mathbf{x}) + \mathbf{P} \quad (6)$$

where $\mathbf{F} \in \Re^{m}$ is the Cartesian-space control input, $H_{\mathbf{x}} \in \Re^{m\times m}$ and $\mathbf{G}_{\mathbf{x}} \in \Re^{m}$ are the Cartesian-space inertia matrix and gravity vector, and where the introduction of $D_{\mathbf{x}} \in \Re^{m\times m^2}$ and $[\dot{\mathbf{x}}\dot{\mathbf{x}}] = [\dot{\mathbf{x}}_1\dot{\mathbf{x}}_1, \cdots, \dot{\mathbf{x}}_1\dot{\mathbf{x}}_m, \ \ \dot{\mathbf{x}}_2\dot{\mathbf{x}}_1, \cdots, \dot{\mathbf{x}}_m\dot{\mathbf{x}}_m]^T \in \Re^{m^2}$ provides an alternative parameterization of the Cartesian-space Coriolis and centripedal acceleration vector. While the form of the Cartesian-space robot dynamics given in (6) is slightly unusual, this representation can be obtained directly from Lagrangian dynamics; for example, the elements of the matrix $D_{\mathbf{x}}$ are the Christoffel symbols (of the first kind) associated with the matrix $H_{\mathbf{x}}$ [18]. The selection of this particular representation (6) is motivated by the work reported for joint-space adaptive unconstrained motion control in [21]. The terms $H_{\mathbf{x}}, D_{\mathbf{x}}, \mathbf{G}_{\mathbf{x}}$ in (6) are bounded functions of $\mathbf{x}$ whose time derivatives $\dot{H}_{\mathbf{x}}, \dot{D}_{\mathbf{x}}, \dot{\mathbf{G}}_{\mathbf{x}}$ depend linearly on $\dot{\mathbf{x}}$. These properties are established in the Appendix and will prove useful in the development of the controller. An additional motivating factor for utilizing the model (6) is, as mentioned above, the desirability of considering the impedance control problem directly in Cartesian-space. Of course, the Cartesian-space control input $\mathbf{F}$ cannot be applied to the robot directly, and must be realized through an equivalent joint torque vector $\mathbf{T}$. Constructing the $\mathbf{F} \rightarrow \mathbf{T}$ map is trivial for nonredundant robots, but represents an interesting problem for robots possessing kinematic redundancy.

It is proposed in this paper that the joint torque $\mathbf{T}$ which ensures that the robot dynamics (6) closely emulates the impedance dynamics (5) can be generated with a controller consisting of three subsystems: a simple filter which characterizes the desired performance (5), a Cartesian-space control scheme which generates the control input $\mathbf{F}$ to the dynamics (6), and an $\mathbf{F} \rightarrow \mathbf{T}$ mapping algorithm which provides the required $\mathbf{T}$ that achieves the desired impedance characteristics and in addition effectively utilizes any available redundancy. A block diagram of this control scheme is given in Figure 1. Observe that the control objective can be realized if the end-effector position $\mathbf{x}$ closely tracks the *desired impedance trajectory* $\mathbf{x}_d \in \Re^m$, defined as the solution to the differential equation:

$$M\ddot{\mathbf{x}}_d + B\dot{\mathbf{x}}_d + N\mathbf{x}_d = -\mathbf{P}_m + M\ddot{\mathbf{x}}_r + B\dot{\mathbf{x}}_r + N\mathbf{x}_r$$
$$\mathbf{x}_d(0) = \mathbf{x}_r(0) \quad , \quad \dot{\mathbf{x}}_d(0) = \dot{\mathbf{x}}_r(0) \qquad (7)$$

obtained from the impedance model (5), where $\mathbf{P}_m$ is the robot force/torque sensor measurement of the contact force $\mathbf{P}$. Equation (7) may be interpreted as defining a simple filter which, given definitions for $\mathbf{x}_r(t), M, B$, and N and on-line measurements $\mathbf{P}_m$, characterizes the desired dynamic relationship between end-effector position and contact force through the specification of $\mathbf{x}_d$. Note that integrating (7) to obtain $\mathbf{x}_d$ requires very little calculation, particularly in the usual case in which M, B, N are diagonal.

The second controller subsystem is an adaptive strategy for generating the control input $\mathbf{F}$ which ensures that the end-effector position $\mathbf{x}$ closely tracks $\mathbf{x}_d$ defined in (7). The proposed controller is a direct adaptive acheme which requires very little information concerning the robot dynamics, and which ensures accurate tracking and global stability in the presence of bounded disturbances. The control law development is influenced by the unconstrained motion controllers proposed in [12,17,19], and also provides an independent derivation of an adaptive law modification related to the e_1-modification strategy proposed for linear time-invarient systems in [20]. The development of this impedance controller is presented in Section 3. The final control subsystem is an algorithm for mapping the Cartesian-space control input to robot joint torque vector. The $\mathbf{F} \rightarrow \mathbf{T}$ map is trivial for nonredundant robots, but proper construction of this map for redundant robots provides a powerful approach to effective redundany utilization [21-25]. A brief summary of the $\mathbf{F} \rightarrow \mathbf{T}$ mapping algorithm is given in Section 4.

3. Adaptive Cartesian Impedance Control Scheme

The function of the Cartesian-space impedance controller is to cause the robot dynamics (6) to evolve in such a way that the end-effector position $\mathbf{x}$ closely tracks $\mathbf{x}_d$ defined in (7). From examination of (7), it can be inferred using measure theory [26] that $\ddot{\mathbf{x}}_d$ is essentially bounded and $\mathbf{x}_d$, $\dot{\mathbf{x}}_d$ are bounded, which is sufficient for our purposes. Define $\mathbf{e} = \mathbf{x}_d - \mathbf{x}$ and $\dot{\mathbf{x}}_d^* = \dot{\mathbf{x}}_d + K\mathbf{e}$, where $K \in \Re^{m \times m}$ is a constant symmetric positive-definite matrix. The term $\dot{\mathbf{x}}_d^*$ is commonly used in adaptive unconstrained motion controllers [e.g., 12,19], and such quantities are also extensively utilized in robot inverse kinematic algorithms [e.g., 27], where these terms are viewed as "modified" desired velocities providing some position error correction.

We now turn to the derivation of the adaptive impedance controller, and note that the development of this scheme is influenced by the work on unconstrained motion control reported in [12,17,19]. Consider the control law

$$\mathbf{F} = A(t)\ddot{\mathbf{x}}_d^* + B(t)[\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*] + \mathbf{f}(t) + \mathbf{P}_m + K_p\mathbf{e} + K_v\dot{\mathbf{e}} \qquad (8)$$

where $K_p, K_v \in \Re^{m \times m}$ are constant symmetric positive-definite matrices and $A(t) \in \Re^{m \times m}$, $B(t) \in \Re^{m \times m^2}$, $\mathbf{f}(t) \in \Re^m$ are adaptive gains whose update laws are to be determined. In this control law, the first four terms on the right side represent feedforward components intended to match elements in the robot dynamics (6), while the last two terms provide simple PD feedback for system stabilization. Observe that, although $A, B, \mathbf{f}$ are clearly the adaptive counterparts to $H_\mathbf{x}, D_\mathbf{x}, \mathbf{G}_\mathbf{x}$ in (6), nothing is assumed regarding the structure of these dynamic model elements except for the dimension of the matrices. Applying the control law (8) to the robot dynamics (6) yields the system error dynamics:

$$H_\mathbf{x}\dot{\mathbf{q}} + V_{cc}\mathbf{q} + \mathbf{\Phi}_H\ddot{\mathbf{x}}_d^* + \mathbf{\Phi}_D[\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*] + \mathbf{\Phi}_G + K_p\mathbf{e} + K_v\dot{\mathbf{e}} + \Delta\mathbf{P} = \mathbf{0} \quad (9)$$

where $\mathbf{q} = \dot{\mathbf{e}} + K\mathbf{e}$ denotes a weighted position-velocity error, $V_{cc} \in \Re^{m \times m}$ is an alternative parameterization for the Coriolis/centripedal acceleration terms (so that $V_{cc}\dot{\mathbf{x}} = D_\mathbf{x}[\dot{\mathbf{x}}\dot{\mathbf{x}}]$), $\mathbf{\Phi}_H = A - H_\mathbf{x}, \mathbf{\Phi}_D = B - D_\mathbf{x}, \mathbf{\Phi}_G = \mathbf{f} - \mathbf{G}_\mathbf{x}$, and $\Delta\mathbf{P} = \mathbf{P}_m - \mathbf{P}$.

The adaptation laws for $A, B, \mathbf{f}$ are now derived using a Lyapunov-stability-based design method. Toward this end, define the Lyapunov function candidate

$$\begin{aligned} V = {} & \frac{1}{2}\mathbf{q}^T H_\mathbf{x}\mathbf{q} + \frac{1}{2}\mathbf{e}^T C_1\mathbf{e} + \frac{1}{2}c_2(\mathbf{\Phi}_G - \delta_1\mathbf{q})^T(\mathbf{\Phi}_G - \delta_1\mathbf{q}) \\ & + \frac{1}{2}\mathrm{tr}[c_3(\mathbf{\Phi}_H - \delta_2\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T)(\mathbf{\Phi}_H - \delta_2\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T)^T \\ & + c_4(\mathbf{\Phi}_D - \delta_3\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T)(\mathbf{\Phi}_D - \delta_3\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T)^T] \end{aligned} \qquad (10)$$

where $C_1 \in \Re^{m \times m}$ is a constant symmetric positive-definite matrix, c_2, c_3, c_4 are positive scalar constants, and $\delta_1, \delta_2, \delta_3$ are nonnegative scalar constants. Note that V in (10) is a positive-definite scalar function of $\mathbf{q}, \mathbf{e}, \mathbf{\Phi}_G, \mathbf{\Phi}_H$, and $\mathbf{\Phi}_D$, and that it is radially unbounded for these arguments. Differentiating V along the error dynamics (9) yields after simplification

$$\begin{aligned} \dot{V} = & -\mathbf{e}^T K^T K_v K\mathbf{e} - \dot{\mathbf{e}}^T K_v\dot{\mathbf{e}} - \delta_1\|\,\mathbf{q}\,\|^2 \\ & - \delta_2\|\,\mathbf{q}\,\|^2\|\,\ddot{\mathbf{x}}_d^*\,\|^2 - \delta_3\|\,\mathbf{q}\,\|^2\|\,\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*\,\|^2 - \mathbf{q}^T\Delta\mathbf{P} \\ & + (\mathbf{\Phi}_G - \delta_1\mathbf{q})^T(c_2\dot{\mathbf{f}} - c_2\dot{\mathbf{G}}_\mathbf{x} - c_2\delta_1\dot{\mathbf{q}} - \mathbf{q}) \\ & + \mathrm{tr}[(\mathbf{\Phi}_H - \delta_2\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T) \\ & \left(c_3\dot{A}^T - c_3\dot{H}_\mathbf{x}^T - c_3\delta_2\frac{d}{dt}(\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T)^T - (\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T)^T\right) \\ & + (\mathbf{\Phi}_D - \delta_3\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T) \\ & \left(c_4\dot{B}^T - c_4\dot{D}_\mathbf{x}^T - c_4\delta_3\frac{d}{dt}(\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T)^T - (\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T)^T\right)] \end{aligned} \qquad (11)$$

where the relationships $K_p = K_v K$ and $C_1 = 2K_p$ are defined, and the identity $\mathbf{q}^T[\dot{H}_\mathbf{x} - 2V_{cc}]\mathbf{q} = 0$ (from Lagrangian dynamics) is utilized [28]. Examination of (11) suggests the following adaptation laws for $\mathbf{f}, A$, and B:

$$\begin{aligned} \dot{\mathbf{f}} &= -\alpha_1\mathbf{f} + \left(\frac{1}{c_2} + \alpha_1\delta_1\right)\mathbf{q} + \delta_1\dot{\mathbf{q}} \\ \dot{A} &= -\alpha_2 A + \left(\frac{1}{c_3} + \alpha_2\delta_2\right)\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T + \delta_2\frac{d}{dt}(\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T) \\ \dot{B} &= -\alpha_3 B + \left(\frac{1}{c_4} + \alpha_3\delta_3\right)\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T + \delta_3\frac{d}{dt}(\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T) \end{aligned} \qquad (12)$$

where the α_i are scalar functions whose form is to be determined. Substituting these adaptation laws into (11) yields

$$\begin{aligned} \dot{V} = & -\mathbf{e}^T K^T K_v K\mathbf{e} - \dot{\mathbf{e}}^T K_v\dot{\mathbf{e}} - \delta_1\|\,\mathbf{q}\,\|^2 - \delta_2\|\,\mathbf{q}\,\|^2\|\,\ddot{\mathbf{x}}_d^*\,\|^2 \\ & - \delta_3\|\,\mathbf{q}\,\|^2\|\,\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*\,\|^2 - \mathbf{q}^T\Delta\mathbf{P} \\ & - \alpha_1 c_2\,\|\,(\mathbf{\Phi}_G - \delta_1\mathbf{q}) + \frac{1}{2\alpha_1}(\dot{\mathbf{G}}_\mathbf{x} + \alpha_1\mathbf{G}_\mathbf{x})\,\|^2 \\ & - \alpha_2 c_3\,\|\,(\mathbf{\Phi}_H - \delta_2\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T) + \frac{1}{2\alpha_2}(\dot{H}_\mathbf{x} + \alpha_2 H_\mathbf{x})\,\|_F^2 \\ & - \alpha_3 c_4\,\|\,(\mathbf{\Phi}_D - \delta_3\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T) + \frac{1}{2\alpha_3}(\dot{D}_\mathbf{x} + \alpha_3 D_\mathbf{x})\,\|_F^2 \end{aligned} \qquad (13)$$

$$+\frac{c_2}{4\alpha_1}\| \dot{\mathbf{G}}_\mathbf{x}+\alpha_1\mathbf{G}_\mathbf{x}\|^2+\frac{c_3}{4\alpha_2}\| \dot{H}_\mathbf{x}+\alpha_2 H_\mathbf{x}\|_F^2$$
$$+\frac{c_4}{4\alpha_3}\| \dot{D}_\mathbf{x}+\alpha_3 D_\mathbf{x}\|_F^2$$

where $\|\cdot\|_F$ denotes the Frobenius norm. Observe that inclusion of σ-modification-like terms in the adaptation laws (12) has led to the appearance of Φ_G, Φ_H, Φ_D in only negative-semidefinite quantities in (13), which in turn will permit the boundedness of $\mathbf{f}, A, B$ to be established [13].

We now examine the positive or potentially positive terms in the expression for $\dot{V}$ given in (13). The following bound is easily established:

$$-\mathbf{q}^T\Delta\mathbf{P} \leq \Delta\mathbf{P}_{max}\|\mathbf{q}\| \tag{14}$$

where $\Delta\mathbf{P}_{max}$ is the scalar upper bound on the (norm of the) force/torque sensor measurement error. The properties of $\mathbf{G}_\mathbf{x}, H_\mathbf{x}$, and $D_\mathbf{x}$ together with the boundedness of $\mathbf{x}_d$ and $\dot{\mathbf{x}}_d$, permit the following bounds to be derived:

$$\begin{aligned}
\frac{1}{\alpha_1}\| \dot{\mathbf{G}}_\mathbf{x}+\alpha_1\mathbf{G}_\mathbf{x}\|^2 &\leq \frac{a_1}{\alpha_1}\|\dot{\mathbf{e}}\|^2+\frac{b_1}{\alpha_1}\|\dot{\mathbf{e}}\| \\
&\quad + c_1\|\dot{\mathbf{e}}\|+\alpha_1 d_1+f_1 \\
\frac{1}{\alpha_2}\| \dot{H}_\mathbf{x}+\alpha_2 H_\mathbf{x}\|_F^2 &\leq \frac{a_2}{\alpha_2}\|\dot{\mathbf{e}}\|^2+\frac{b_2}{\alpha_2}\|\dot{\mathbf{e}}\| \\
&\quad + c_2\|\dot{\mathbf{e}}\|+\alpha_2 d_2+f_2 \\
\frac{1}{\alpha_3}\| \dot{D}_\mathbf{x}+\alpha_3 D_\mathbf{x}\|_F^2 &\leq \frac{a_3}{\alpha_3}\|\dot{\mathbf{e}}\|^2+\frac{b_3}{\alpha_3}\|\dot{\mathbf{e}}\| \\
&\quad + c_3\|\dot{\mathbf{e}}\|+\alpha_3 d_3+f_3
\end{aligned} \tag{15}$$

for some positive scalar constants a_i, b_i, c_i, d_i, f_i. It can be seen from (14)-(15) that the positive or potentially positive terms in $\dot{V}$ are at most quadratically dependent on $\|\mathbf{e}\|$, $\|\dot{\mathbf{e}}\|$, while the negative quantities in $\dot{V}$ depend on higher order polynomials in these arguments; this fact could be used to prove the global stability of the proposed controller. However, we can improve the situation, and the performance of the controller, by reducing the dependence of each of the terms in (15) to at most a linear one in $\|\dot{\mathbf{e}}\|$. This can be achieved by defining

$$\alpha_i=\gamma_i\|\dot{\mathbf{e}}\|^2 \qquad i=1,2,3 \tag{16}$$

where the γ_i are positive constants. The bounds (14),(15) and the definitions (16) permit (13) to be written

$$\begin{aligned}
\dot{V} \leq & -\mathbf{e}^T K^T K_v K\mathbf{e}-\dot{\mathbf{e}}^T K_v\dot{\mathbf{e}}-\delta_1\|\mathbf{q}\|^2 \\
& -\delta_2\|\mathbf{q}\|^2\|\ddot{\mathbf{x}}_d^*\|^2-\delta_3\|\mathbf{q}\|^2\|\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*\|^2 \\
& -\gamma_1 c_2\|\dot{\mathbf{e}}\|\cdot\|\Phi_G-\delta_1\mathbf{q} \\
& \qquad +\frac{1}{2\gamma_1\|\dot{\mathbf{e}}\|}(\dot{\mathbf{G}}_\mathbf{x}+\gamma_1\|\dot{\mathbf{e}}\|\mathbf{G}_\mathbf{x})\|^2 \\
& -\gamma_2 c_3\|\dot{\mathbf{e}}\|\cdot\|\Phi_H-\delta_2\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T \\
& \qquad +\frac{1}{2\gamma_2\|\dot{\mathbf{e}}\|}(\dot{H}_\mathbf{x}+\gamma_2\|\dot{\mathbf{e}}\|H_\mathbf{x})\|_F^2 \\
& -\gamma_3 c_4\|\dot{\mathbf{e}}\|\cdot\|\Phi_D-\delta_3\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T \\
& +\frac{1}{2\gamma_3\|\dot{\mathbf{e}}\|}(\dot{D}_\mathbf{x}+\gamma_3\|\dot{\mathbf{e}}\|D_\mathbf{x})\|_F^2 \\
& +\Delta\mathbf{P}_{max}\|\mathbf{q}\| + \eta_0 + \eta_1\|\dot{\mathbf{e}}\|
\end{aligned} \tag{17}$$

where η_0, η_1, are positive scalar constants. Introducing the definitions (16) and $\beta_i = 1/c_{i+1}$ for $i=1,2,3$ in (12) yields the following form for the adaptation laws:

$$\dot{\mathbf{f}}=-\gamma_1\|\dot{\mathbf{e}}\|\mathbf{f}+(\beta_1+\delta_1\gamma_1\|\dot{\mathbf{e}}\|)\mathbf{q}+\delta_1\dot{\mathbf{q}}$$

$$\begin{aligned}
\dot{A} =& -\gamma_2\|\dot{\mathbf{e}}\|A+(\beta_2+\delta_2\gamma_2\|\dot{\mathbf{e}}\|)\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T \\
& +\delta_2\frac{d}{dt}(\mathbf{q}(\ddot{\mathbf{x}}_d^*)^T) \\
\dot{B} =& -\gamma_3\|\dot{\mathbf{e}}\|B+(\beta_3+\delta_3\gamma_3\|\dot{\mathbf{e}}\|)\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T \\
& +\delta_3\frac{d}{dt}(\mathbf{q}(\dot{\mathbf{x}}\dot{\mathbf{x}}_d^*)^T)
\end{aligned} \tag{18}$$

Careful study of (17) reveals that the region in which $\dot{V} \geq 0$ forms a compact set for $\mathbf{e}, \dot{\mathbf{e}}, \Phi_G, \Phi_H, \Phi_D$ that includes the origin, from which it follows that all of these quantities are bounded [13]. This makes sense because the positive terms in (17) depend linearly on $\|\mathbf{e}\|$, $\|\dot{\mathbf{e}}\|$ while the negative terms depend on higher order polynomials of $\|\mathbf{e}\|$, $\|\dot{\mathbf{e}}\|$. Thus the rate of growth of the negative terms is much faster than the positive terms, and the impedance tracking errors must ultimately converge to a compact set. The fact that the negative terms in (17) include quantities proportional to $\|\mathbf{e}\|^4$ suggests that the bounds on the tracking errors are small.

Several observations can be made concerning the direct adaptive impedance controller (8),(18). First, the preceding analysis demonstrates that this scheme ensures that the robot dynamics (6) tracks $\mathbf{x}_d$ defined in (7) with global stability. Inspection of (17) implies that the size of the tracking error can be made arbitrarily small by increasing the controller gains. Of course, the above analysis assumes continuous-time implementation, so that in actual computer control applications consideration of discretization effects would prohibit the use of excessively large gains. Computer simulation results [29] have indicated that excellent tracking is achieved with the proposed controller using moderate sampling rates (on the order of 100Hz), and that this tracking is relatively insensitive to the selection of controller gains. A second observation regarding the control scheme (8),(18) is that the adaptation laws (18) include both proportional and integral adaptive elements. Note that proportional-plus-integral adaptation laws provide superior transient response, improved convergence, and increased flexibility compared with integral adaptation laws [17]. Of course, integral adaptation is easily recovered from (18) by setting $\delta_i = 0$. Note also that in (18), the scalar adaptation gains $\beta_i, \gamma_i, \delta_i$ can be replaced with the appropriate diagonal matrix gains for increased flexibility; scalar gains were used in the derivation of (18) for simplicity of development.

Finally, it is interesting to note the relationship between the adaptation laws (18) and the e_1-modification strategy proposed for linear time-invariant systems [20]. Very briefly, the e_1-modification adaptation laws are similar to those of the well-known σ-modification scheme, except that the σ term in the update laws is replaced with a term proportional to the system output error $e_1 \in \Re$. This refinement yields an adaptive controller with robustness properties similar to the σ-modification scheme, but often with improved convergence characteristics. The adaptation laws (18) contain a similar term, with e_1 replaced by $\|\dot{\mathbf{e}}\|$, and preliminary study has indicated similar advantages. For example, a linearization analysis of the controller (8),(18) shows that, in the case where the dynamic model may be considered "slowly varying" and the force/torque sensor measurement error is negligible, the stability of the controller becomes asymptotic stability. Further study of the convergence properties of the controller (8),(18) is currently underway.

In summary, the direct adaptive impedance control scheme (8),(18) ensures that the robot dynamics (6) evolves so that $\mathbf{x}$ tracks $\mathbf{x}_d$ closely and with global stability. The controller has a simple structure and is suitable for on-line implementation with relatively modest computing power. The control input $\mathbf{F}$ is calculated entirely based on the observed performance of the robot, so that knowledge of the structure or parameter values of the robot dynamics is not required. No assumptions are made concerning the behavior of the robot dynamics in demonstrating global stability; however, in the event the the dynamic model terms $H_{\mathbf{x}}, D_{\mathbf{x}}, \mathbf{G}_{\mathbf{x}}$ vary slowly relative to the adaptive elements and the force/torque sensor measurement error is negligible, the proposed scheme provides asymptotic stability in which $\mathbf{x} \rightarrow \mathbf{x}_d$ as $t \rightarrow \infty$ [29].

4. $\mathbf{F} \rightarrow \mathbf{T}$ Mapping Algorithm

Observe that the Cartesian-space control input $\mathbf{F} \in \Re^m$ computed in (8),(18) cannot physically be applied to the robot end-effector; therefore, this desired control input must be mapped into an equivalent joint torque vector $\mathbf{T} \in \Re^n$. For nonredundant robots ($m = n$), this $\mathbf{F} \rightarrow \mathbf{T}$ map is uniquely defined and is given by [e.g., 28]

$$\mathbf{T} = J^T(\theta)\mathbf{F} \tag{19}$$

For kinematically redundant robots ($m < n$), the $\mathbf{F} \rightarrow \mathbf{T}$ mapping problem is underdetermined, so that there exists an infinite number of joint torque vectors corresponding to a given control input $\mathbf{F}$. By properly selecting a particular joint torque vector from among the possible candidates, the robot redundancy can be utilized to improve some measure of task performance. Resolving the redundancy through the construction of an appropriate $\mathbf{F} \rightarrow \mathbf{T}$ map represents a departure from the conventional approach in which the redundancy is resolved at the inverse kinematics level; a general and comprehensive discussion of this approach to redundant robot control may be found in [21-25]. It is interesting to observe that effective redundancy resolution can be important with conventional six degree-of-freedom manipulators when these robots are involved in applications with a variable dimensional task-space. For example, an operation may require full end-effector orientation capabilities for part of the task and only end-effector "pointing" for the remainder of the operation, so that the dimension of the task-space changes from $m = 6$ to $m = 5$. Note that the robot possesses one degree of redundancy during the "pointing" portion of this operation. In this section, the problem of constructing on appropriate $\mathbf{F} \rightarrow \mathbf{T}$ map is briefly considered for completeness.

Recall that for a redundant robot, $\mathbf{F} \in \Re^m$ and $\mathbf{T} \in \Re^n$ with $m < n$. The problem of constructing an appropriate $\mathbf{F} \rightarrow \mathbf{T}$ map may be formulated in terms of inverting the known $\mathbf{T} \rightarrow \mathbf{F}$ map, which is unique even for redundant robots. The $\mathbf{T} \rightarrow \mathbf{F}$ has the form [21]

$$\mathbf{F} = M(\theta)\mathbf{T} \tag{20}$$

where it can be verified that $M = (JH^{-1}J^T)^{-1}JH^{-1} \in \Re^{m \times n}$. Inversion of the $\mathbf{T} \rightarrow \mathbf{F}$ map (24) may be achieved in two ways: "direct" inversion using the theory of generalized inverses, or "indirect" inversion in which both M and $\mathbf{F}$ are augmented with $r = n - m$ additional rows and the resulting fully-determined system is inverted utilizing standard methods. The direct approach to inverting (20) is useful for realizing dynamic performance objectives, primarily because inverting (20) using generalized inverse theory readily permits optimization of objective functions involving joint torque $\mathbf{T}$ and joint acceleration $\ddot{\theta}$. For example, the problem of globally minimizing robot kinetic energy subject to the constraint that the end-effector motion $\mathbf{x}(t)$ closely tracks the desired impedance trajectory $\mathbf{x}_d(t)$ can be solved in this manner.

The indirect approach to inverting the $\mathbf{T} \rightarrow \mathbf{F}$ map (20) is useful primarily for realizing kinematic performance objectives. The idea of augmenting M and $\mathbf{F}$ with r additional rows to obtain a fully-determined system is conceptually simple, and can be accomplished by augmenting the vector of end-effector generalized coordinates $\mathbf{x}$ with an additional generalized coordinate vector $\psi \in \Re^r$ [22]. The vector ψ is chosen so that the robot redundancy is resolved in some useful manner. Many kinematic performance objectives can be quantified by specifying that, in addition to performing the desired m-dimensional end-effector task, the robot must perform an r-dimensional additional task defined by the following kinematic relationship [22]:

$$\mathbf{g}(\theta) = \psi_r(t) \tag{21}$$

where $\mathbf{g} : \Re^n \rightarrow \Re^r$ is a kinematic function whose desired evolution is defined by $\psi_r(t) \in \Re^r$. For example, this strategy can be utilized to optimize general kinematic performance objectives subject to the constraint of realizing the end-effector task.

In summary, the proposed impedance control algorithm for manipulators shown in Figure 1 consists of the simple filter (7), control law (8), and adaptation laws (18), together with the appropriate $\mathbf{F} \rightarrow \mathbf{T}$ map. The filter (7) characterizes the desired impedance relationship between end-effector error and contact force, the Cartesian-space impedance controller (8),(18) ensures that this impedance is realized, and the $\mathbf{F} \rightarrow \mathbf{T}$ map can be chosen to effectively utilize any available kinematic redundancy to improve robot performance. Note that this control scheme can be applied directly to trajectory tracking in free-space motion by removing the impedance filter.

5. Conclusions

This paper presents a direct adaptive impedance control strategy for robot manipulators. The controller does not require knowledge of either the structure or the parameter values of the robot dynamics, and is implemented without calculation of the robot inverse kinematic transformation. As a result, the scheme is very general and computationally efficient. It is shown that the controller ensures accurate emulation of the desired impedance model, and provides global stability in the presence of bounded disturbances.

Future research will involve a deeper analysis of the convergence properties of the control strategy, including a quantitative study of the effects of controller gains on the size of the residual error. Additionally, applications of the proposed control scheme to operations in remote, harsh, and unstructured surroundings will be addressed, as well as the effective utilization of redundancy to improve performance in such environments.

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7. References

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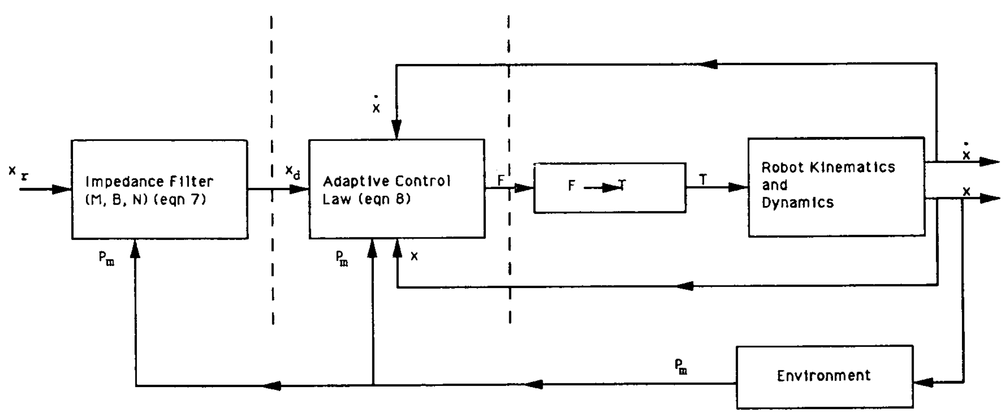

Figure 1: Impedance Control Block Diagram